AWS Lab: S3 Versioning Activation & Accidental Deletion Rollback

Duration : 30 Min.

Scenario

A deployment script accidentally overwrote and deleted a critical application configuration file 'app-config.json' in an S3 bucket. Fortunately, S3 Versioning had been enabled. The engineer must inspect the bucket's object versions, locate the Delete Marker, remove the Delete Marker to restore the object, and retrieve the prior version.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon S3 Object Versioning - **Target File:** `app-config.json` - **Remediation Action:** Delete active Delete Marker to rollback to prior version

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Check bucket versioning status using aws s3api get-bucket-versioning. 2. List object versions and delete markers using aws s3api list-object-versions. 3. Delete the active Delete Marker to restore the latest non-deleted object version. 4. Verify that the restored object matches the original MD5 checksum, saving commands to ~/s3_version_recovery.sh.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Versioning query: verify bucket versioning configuration is active	3 Marks
TC2	Version enumeration: list-object-versions identifies the specific Delete Marker version ID	2 Marks
TC3	Object restoration: delete-object removes Delete Marker restoring the underlying version	3 Marks
TC4	Checksum validation: verify restored object content matches expected baseline	2 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.